# Assessment of the 16 Post-Processing Approaches

What was actually tested, what was only partially screened, and what remains open

## Bottom line

No: the 16 approaches were not all implemented and invalidated. The audit log itself says that only a small subset was tested directly, several were only partially tested, and some were merely demoted because a shared precondition looked weak. That distinction matters, because a failed proxy or precondition is not equivalent to a failed method.

| # | Approach | Status | What the evidence actually says |
|---|---|---|---|
| 1 | Geodesic / shortest-path completion | **Partially screened** | exp109 used mean learned foreground in a 35 degree cone ahead of endpoints. exp110 improved the audit by examining best rays. Learned semantic support remained extremely weak: best-ray mean was 0.0082 for truncated masks versus 0.0137 for healthy masks, and the minimum along rays was 0 for both. However, raw-image dark absorption achieved AUC 0.611, so image-derived geodesics remain open. |
| 2 | Elastic rod / Euler-elastica | **Untested** | No implementation or direct test. Its usefulness depends on whether any usable continuation evidence exists, but that dependency is not itself a test of elastica. |
| 3 | Level sets / active contours | **Untested** | No direct implementation. The axial/lateral error decomposition only bears on likely scope: boundary evolution may help lateral under-segmentation but cannot by itself recover long missing distal extent unless the evolution region permits it. |
| 4 | CRF / graph-cut refinement | **Demoted, not refuted** | exp108 found about 59% of the missing area is axial, beyond mask ends, and about 41% lateral. That weakens narrow-band graph cuts as a universal repair but still leaves a substantial lateral-error population. |
| 5 | Vesselness / ridge filtering | **Partially tested** | exp103 tested Hessian ridge filtering as a global proposal source. Retaining 80% of filament pixels required accepting 26.2% of the solar disk. Local ridge evidence specifically ahead of endpoints was not tested. |
| 6 | Orientation fields | **Untested** | No direct orientation-field audit appears in the usable log. This remains especially relevant as a directional gate for endpoint continuation. |
| 7 | Persistent homology | **Untested** | No direct experiment. It remains a possible way to stabilize skeleton topology or reject short-lived branches, but current evidence does not test it. |
| 8 | Minimum-cost flow / global assignment | **Untested; low ceiling** | Not implemented. Fragmentation is only a small subset: 29 components, with an oracle around 14 rescues. That limits expected upside but does not invalidate the method. |
| 9 | Matched filtering | **Untested** | Explicitly never attempted. |
| 10 | Anisotropic diffusion | **Untested** | No actual experiment documented. It remains a preprocessing option for strengthening orientation or ridge evidence. |
| 11 | Distance-transform / marker-controlled watershed | **Untested as proposed** | The log indicates that a watershed using a different gate remains untested. The split oracle motivates it, but no actual gated watershed result is reported. |
| 12 | Medial-axis width profile | **Partially tested** | Two scalar summaries were tested: width CV gave AUC 0.457; rho = w_min / median(w) gave AUC 0.712 but only about 0.067 precision at the useful operating point. Richer structure-preserving statistics such as saddle depth, slopes, peak ratios, and multimodality were only queued. |
| 13 | Shape derivatives / sensitivity analysis | **Untested** | No attempt reported. |
| 14 | Optimal transport between predictions | **Untested** | exp107 tested plain rasterized cross-model disagreement instead, with AUC around 0.595/0.574. That is not optimal transport. |
| 15 | Test-time energy minimization | **Untested** | Deliberately deferred. It remains a possible unifying framework after individual evidence terms prove useful. |
| 16 | Endpoint geodesic + orientation support | **Precondition only** | Only the semantic-path side was screened. Orientation support itself was not properly evaluated, and raw forward absorption did carry signal. The combined method therefore remains open. |

## Important technical correction on the geodesic rejection

The claim that a geodesic is broken because every candidate ray contains at least one pixel with zero learned support is not generally valid. If the path cost is finite, for example:

```
c(x) = alpha * (1 - P(x)) + ...
```

then P(x) = 0 merely gives that pixel a high finite cost. Dijkstra, fast marching, or another shortest-path method can still cross it if competing paths are worse. A zero-support pixel blocks the path only if the implementation explicitly assigns infinite cost or forbids traversal.

A proper geodesic audit should therefore compute an actual minimum path cost, not only best-ray summary statistics:

```
C*(p) = min over paths gamma from p to boundary(R) of sum_{x in gamma} c(x)
```

Then compare minimum path cost, path-integrated dark absorption, orientation consistency, and endpoint plausibility between truncated and healthy instances.

## Why dark_best_ray matters

The AUC of 0.611 is not strong enough to be a practical classifier by itself; the reported best precision is only about 0.157 and the operating point is net-negative. But it is scientifically important because it shows that the raw image contains forward information that the learned semantic field did not capture. That argues against declaring endpoint continuation dead.

## Why the width-profile result is still open

rho = w_min / median(w) reached AUC 0.712 but had poor precision. That means the feature carries distributional signal but lacks specificity. A richer local profile can preserve structure that one scalar destroys. Useful follow-ups include saddle depth, local derivatives dw/ds, peak prominence, symmetric width expansion around a candidate neck, and multimodality of w(s).

## Correct current accounting

**Clearly weakened:** semantic-only endpoint support; global Hessian-ridge proposals; simple single-statistic neck detection.

**Partially investigated:** geodesic continuation; vesselness; medial-axis analysis; cross-model disagreement.

**Demoted because the relevant failure population is small:** global fragment linking / minimum-cost assignment.

**Essentially untouched:** elastica; active contours; graph cuts as an actual implementation; orientation fields; persistent homology; matched filtering; anisotropic diffusion; marker-controlled watershed; shape derivatives; optimal transport; test-time energy minimization.

## Most defensible next direction

The strongest immediate follow-up is not another semantic-support audit. The interesting result is that raw forward absorption carries more information than the learned foreground field. The next experiment should therefore combine raw-image forward evidence with directional geometry: orientation or ridge consistency along a candidate continuation path. That directly tests whether faint distal filament extent is recoverable from image structure even when the semantic model is silent.

Source basis: the attached experiment/audit log supplied in this conversation. This document preserves the distinction between direct tests, partial tests, and precondition screens rather than treating them as equivalent.